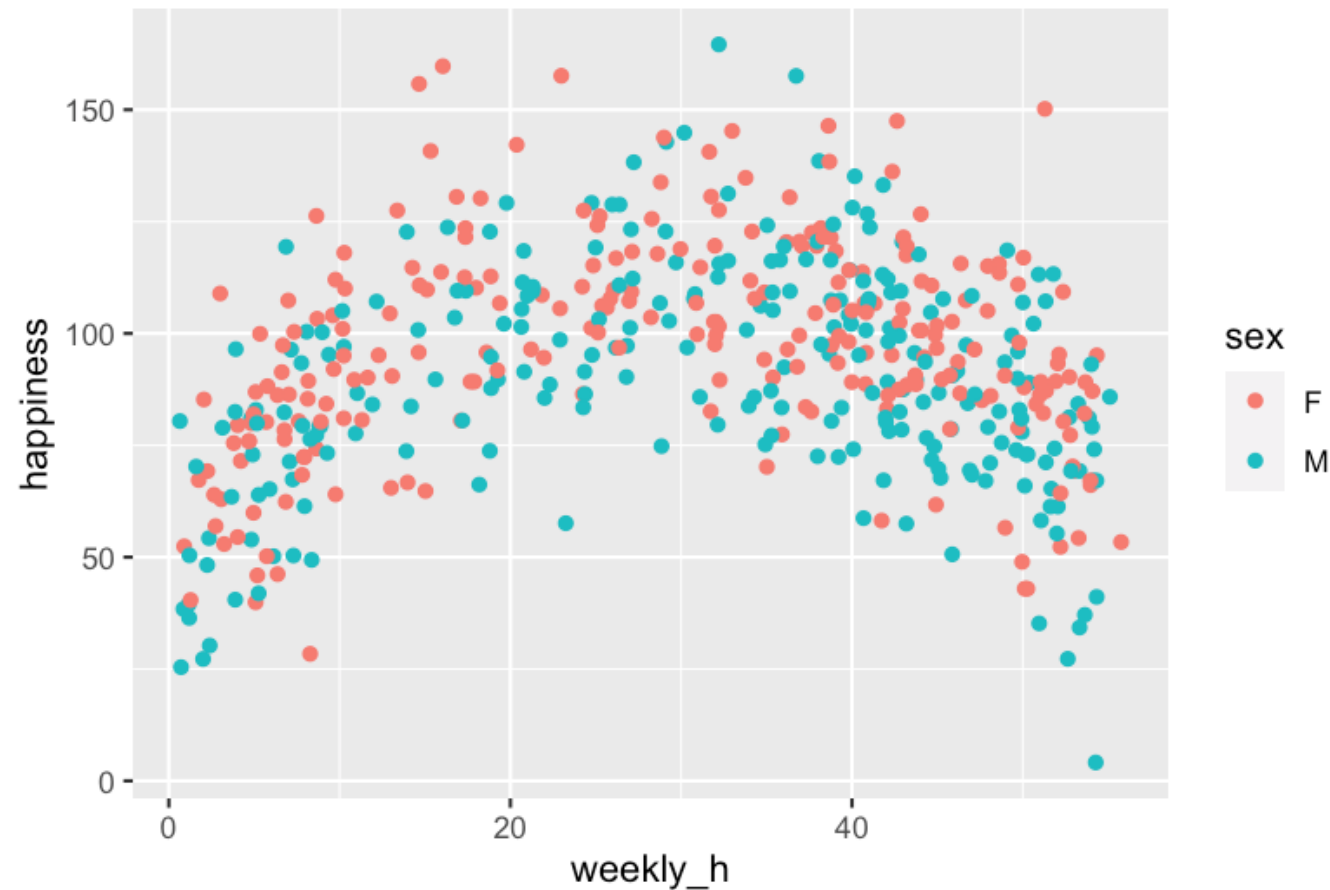


Non-linear relationship: happiness vs. working hours

To assess the relationship between happiness and working hours, a sample of 498 individuals is considered.



Non-linear relationship: happiness vs. working hours

Would you use a linear regression model to predict the level of happiness according to the number of hours worked and sex at birth in the reference population?

```
> summary(mod_lin)
```

Call:

```
lm(formula = happiness ~ weekly_h + sex, data = happy)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-89.462	-14.059	-0.489	16.494	74.942

Coefficients:

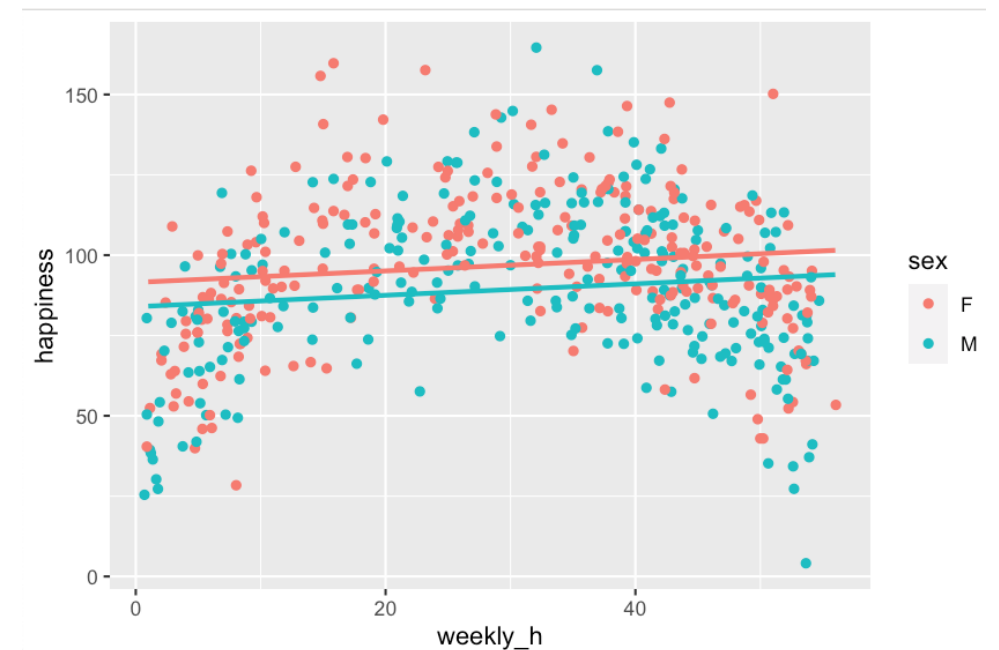
	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	91.52615	2.48441	36.840	< 2e-16 ***
weekly_h	0.17825	0.06662	2.676	0.007707 **
sexM	-7.57482	2.19407	-3.452	0.000603 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 24.42 on 495 degrees of freedom

Multiple R-squared: 0.03479, Adjusted R-squared: 0.03089

F-statistic: 8.92 on 2 and 495 DF, p-value: 0.0001564



Non-linear relationship: happiness vs. working hours

Quadratic Model: $Y = \beta_0 + \beta_1 * weekly_h + \beta_2 * weekly_h^2 + \beta_3 * I(sex = M) + \epsilon$

```
> mod_quad<-lm(happiness~weekly_h+weekly_h2+sex,data=happy)
> summary(mod_quad)
```

Call:

```
lm(formula = happiness ~ weekly_h + weekly_h2 + sex, data = happy)
```

Residuals:

Min	1Q	Median	3Q	Max
-61.954	-12.643	-0.011	12.294	67.104

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	56.591328	2.783248	20.333	< 2e-16 ***
weekly_h	4.034959	0.225388	17.902	< 2e-16 ***
weekly_h2	-0.068929	0.003918	-17.591	< 2e-16 ***
sexM	-7.412487	1.722175	-4.304	2.02e-05 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 19.17 on 494 degrees of freedom

Multiple R-squared: 0.4065, Adjusted R-squared: 0.4029

F-statistic: 112.8 on 3 and 494 DF, p-value: < 2.2e-16

